Exercise 1-6

Exercise 1

a) $\text{Mean} = \dfrac{8+7+4+5+9+13+10+8+8+7+6+5+3+11+10+8+5+4+8+6}{20}$

$= \dfrac{145}{20}$

$= 7.25$

b) $\cancel{3}, \cancel{4}, \cancel{4}, \cancel{5}, \cancel{5}, \cancel{5}, \cancel{6}, \cancel{6}, \cancel{7}, 7, 8, \cancel{8}, \cancel{8}, \cancel{8}, \cancel{8}, \cancel{9}, \cancel{10}, \cancel{10}, \cancel{11}, \cancel{13}$

$\text{Median} = \dfrac{7+8}{2}$

$= \dfrac{15}{2}$

$= 7.5$

c) Mode $= 8$

Exercise 2

a) $\text{Mean} = \dfrac{2001(4)+2002(5)+2003(4)+2004(2)+2005(10)+2006(5)+2007(3)+2008(5)}{4+5+4+2+10+5+3+5}$

$= \dfrac{76175}{38}$

$= 2004.61$

$\cancel{2001}, \cancel{2001}, \cancel{2001}, \cancel{2001}, \cancel{2002}, \cancel{2002}, \cancel{2002}, \cancel{2002}, \cancel{2002}, \cancel{2003}, \cancel{2003}, \cancel{2003}, \cancel{2003}, \cancel{2004},$
$\cancel{2004}, \cancel{2005}, \cancel{2005}, \cancel{2005}, 2005, 2005, \cancel{2005}, \cancel{2005}, \cancel{2005}, \cancel{2005}, \cancel{2005}, \cancel{2006}, \cancel{2006}, \cancel{2006},$
$\cancel{2006}, \cancel{2006}, \cancel{2007}, \cancel{2007}, \cancel{2007}, \cancel{2008}, \cancel{2008}, \cancel{2008}, \cancel{2008}, \cancel{2008}$

$\text{Median} = \dfrac{2005+2005}{2}$

$= 2005$

Mode $= 2005$

b) Mode

c) Mean

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Exercise 3

a)

Length (mm)	Midpoint	Frequency	Cumulative Frequency
149.5 - 154.5	152	5	5
154.5 - 159.5	157	2	7
159.5 - 164.5	162	6	13
164.5 - 169.5	167	8	21
169.5 - 174.5	172	9	30
174.5 - 179.5	177	11	41
179.5 - 184.5	182	6	47
184.5 - 189.5	187	3	50

$$\text{Mean} = \frac{152(5)+157(2)+162(6)+167(8)+172(9)+177(11)+182(6)+187(3)}{5+2+6+8+9+11+6+3}$$

$$= \frac{8530}{50}$$

$$= 170.6$$

b) $\frac{N}{2} = \frac{50}{2}$ $\quad\therefore$ Median class = 169.5 - 174.5

$= 25$

$$\text{Median} = L + \frac{\frac{N}{2} - cf_p}{f_{med}}(w)$$

$$= 169.5 + \frac{25-21}{9}(5)$$

$$= 171.72$$

c) $$\text{Mode} = l + h \times \frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)}$$

$$= 174.5 + 5 \times \frac{11-9}{2(11)-9-6}$$

$$= 175.93$$

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Exercise 4

a) Sample of students' cars

Age	Midpoint	Frequency	Cumulative Frequency
0.5 - 3.5	2	23	23
3.5 - 6.5	5	33	56
6.5 - 9.5	8	63	119
9.5 - 12.5	11	68	187
12.5 - 15.5	14	19	206
15.5 - 18.5	17	10	216
18.5 - 21.5	20	1	217
21.5 - 24.5	23	0	217

$$\text{Mean} = \frac{2(23)+5(33)+8(63)+11(68)+14(19)+17(10)+20(1)+23(0)}{23+33+63+68+19+10+1+0}$$

$$= \frac{1919}{217}$$

$$= 8.843$$

$$\frac{N}{2} = \frac{217}{2}$$

$$= 108.5$$

$\therefore$ Median class $= 6.5 - 9.5$

$$\text{Median} = L + \frac{\frac{N}{2} - cf}{f_{med}}(w)$$

$$= 6.5 + \frac{108.5 - 56}{63}(3)$$

$$= 9.0$$

$$\text{Mode} = l + h \times \frac{f_1 - f_0}{2f_1 - f_0 - f_2}$$

$$= 9.5 + 3 \times \frac{68 - 63}{2(68) - 63 - 19}$$

$$= 9.778$$

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b) Sample of staffs' cars

Age	Midpoint	Frequency	Cumulative Frequency
0.5-3.5	2	30	30
3.5-6.5	5	47	77
6.5-9.5	8	36	113
9.5-12.5	11	30	143
12.5-15.5	14	8	151
15.5-18.5	17	0	151
18.5-21.5	20	0	151
21.5-24.5	23	1	152

$$\text{Mean} = \frac{2(30)+5(47)+8(36)+11(30)+14(8)+17(0)+20(0)+23(1)}{152}$$

$$= \frac{1048}{152}$$

$$= 6.895$$

$$\frac{N}{2} = \frac{152}{2}$$

$$= 76$$

$\therefore$ Median class = 3.5-6.5

$$\text{Median} = L + \frac{\frac{N}{2} - cf_p}{f_{med}}(w)$$

$$= 3.5 + \frac{76-30}{47}(3)$$

$$= 6.436$$

$$\text{Mode} = l + h \times \frac{f_1 - f_0}{2f_1 - f_0 - f_2}$$

$$= 3.5 + 3 \times \frac{47-30}{2(47)-30-36}$$

$$= 5.321$$

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Exercise 5

a) i) $Y[k] = 0.8143$

$\frac{\text{Number of values less than } 0.8143}{\text{Total number of values}}(100) = i$

$i = \frac{8}{18}(100)$

$i = 44.4$

$k = 45$

0.8143 is 45^{th} percentile.

ii) $Y[k] = 0.8062$

$i = \frac{\text{Number of values less than } 0.8062}{\text{Total number of values}}(100)$

$i = \frac{2}{18}(100)$

$i = 11.11$

$k = 12$

0.8062 is 12^{th} percentile.

b) i) $i = \frac{80}{100}(18)$

$i = 14.4$

$k = 15$

$P_{80} = Y[15]$

$= 0.8161$

ii) $Q_3 = P_{75}$

$i = \frac{75}{100}(18)$

$i = 13.5$

$k = 14$

$Q_3 = Y[14]$

$= 0.8161$

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iii) $i = \frac{33}{100}(18)$

$i = 5.94$

$k = 6$

$P_{33} = Y[6]$
$= 0.8110$

iv) $Q_1 = P_{25}$

$i = \frac{25}{100}(18)$

$i = 4.5$

$k = 5$

$Q_1 = Y[5]$
$= 0.8079$

Exercise 6

a) Range $= 56 - 34$
$= 22$

b) $\bar{x} = \frac{45+49+42+56+41+36+34+38+41+45+40+42+41+39+38+40+39+36+4}{19}$

$\bar{x} = \frac{783}{19}$

$\bar{x} = 41.21$

$s^2 = \frac{\Sigma(x_i - \bar{x})^2}{n-1}$

$s^2 = \frac{2(45-41.21)^2 + (49-41.21)^2 + 2(42-41.21)^2 + (56-41.21)^2 + 4(41-41.21)^2 + 2(36-41.21)^2 + (34-41.21)^2 + 2(38-41.21)^2 + 2(40-41.21)^2 + 2(39-41.21)^2}{19-1}$

$s^2 = \frac{449.1579}{18}$

$s^2 = 24.953$

c) $s = \sqrt{24.953}$

$s = 4.9953$